

CHAPTER 08

Classification Part 2: Linear Classifiers & SVMs

Drawing Lines & Maximizing Margins

► CAI1001C | Artificial Intelligence
Thinking

► Professor Carlos
Marquez

► Miami Dade
College

Learning Objectives

WHAT WE'LL MASTER TODAY



Train a **Logistic Regression** classifier and interpret its probabilities.



Train a **Support Vector Machine (SVM)** and explain margin maximization.



Read a **Classification Report** to distinguish Precision from Recall.



Compare four classifiers side-by-side to make a **reasoned judgment**.

Logistic Regression

THE DECISION LINE

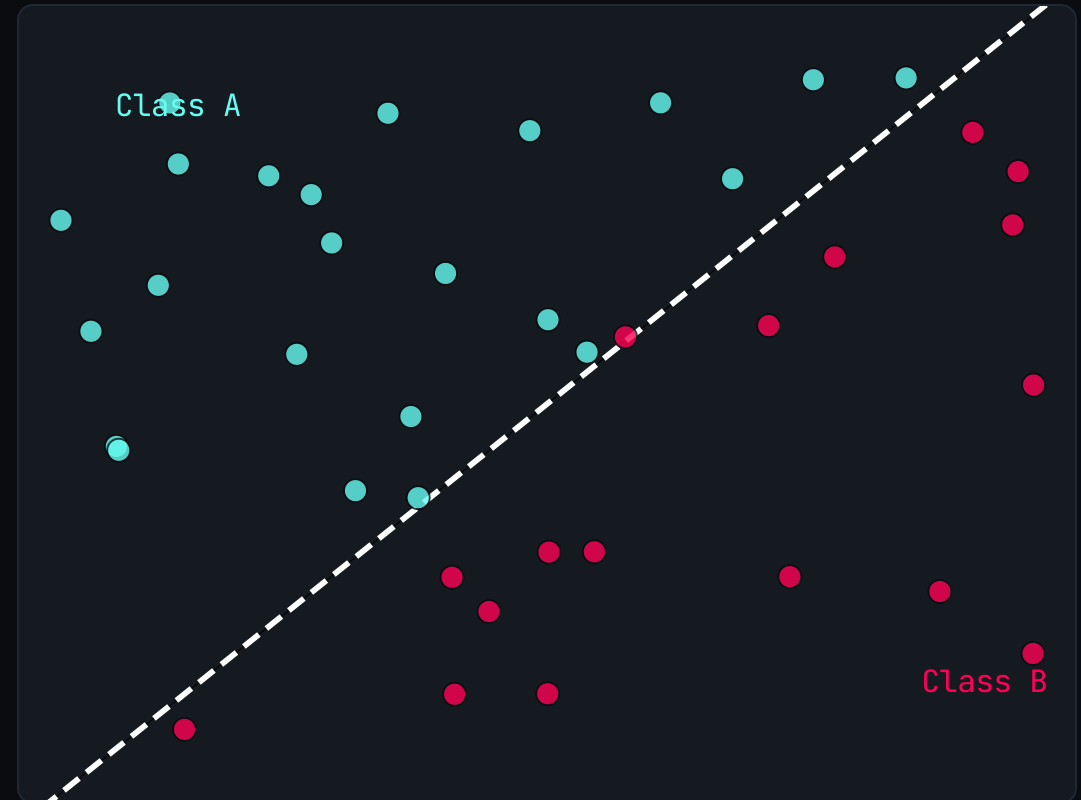
Despite the name, it is a **classification** algorithm, not regression.

How it works:

1. Assigns weights to features.
2. Calculates a score.
3. Converts score to a **probability**.
4. Draws a line (decision boundary) at 0.5.

⚠ Common Pitfall:

"Regression" refers to the math underneath, but the output is a class label (e.g., Spam or Not Spam).



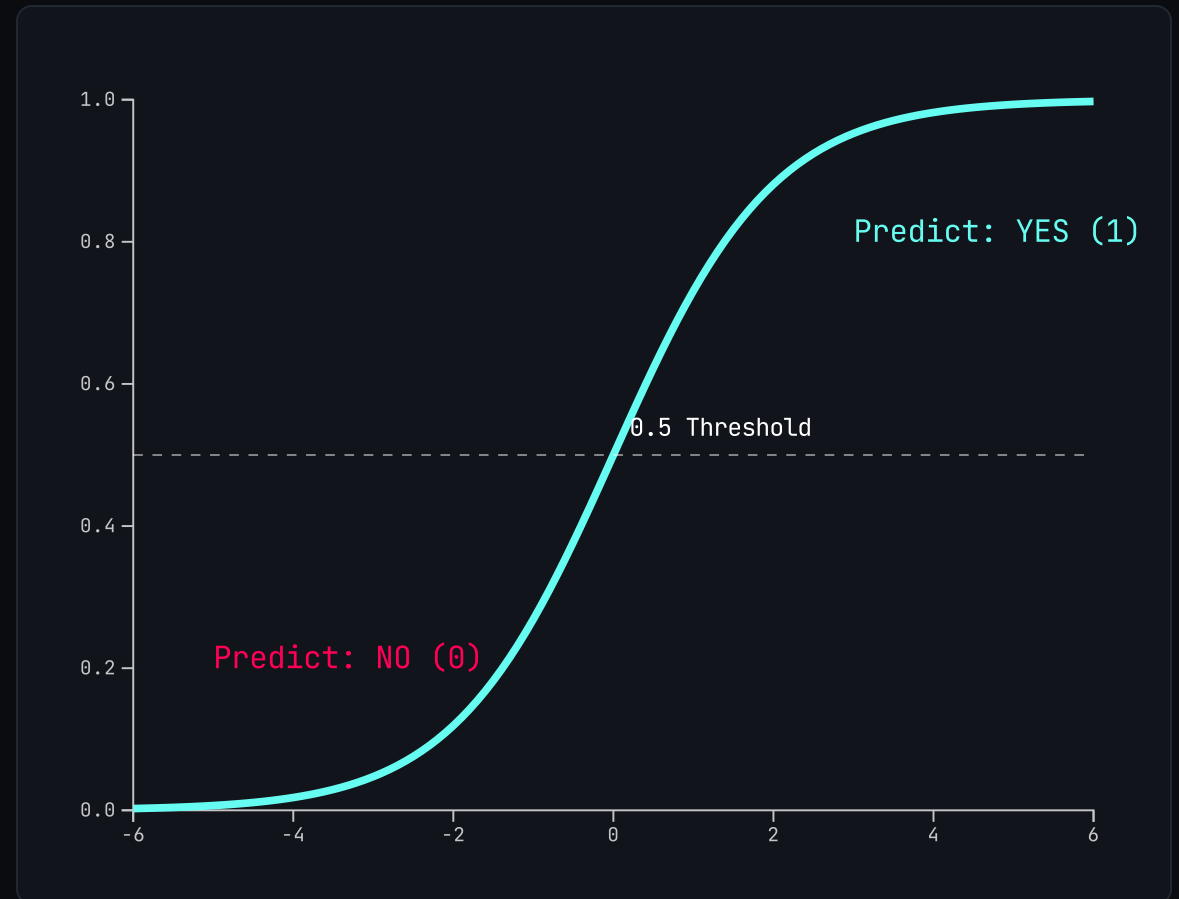
The Sigmoid Function

SQUASHING SCORES INTO PROBABILITIES

How do we turn a raw score (like 532.7) into a probability (0 to 1)?

$$P(y=1) = 1 / (1 + e^{-z})$$

- **Input:** Any number $(-\infty \text{ to } +\infty)$
- **Output:** Strictly between 0 and 1
- **Threshold:** At 0.5, the decision flips



Support Vector Machines

MAXIMIZING THE MARGIN

Logistic Regression finds a line. SVM finds the **widest possible street** between the classes.

COMPONENT 01

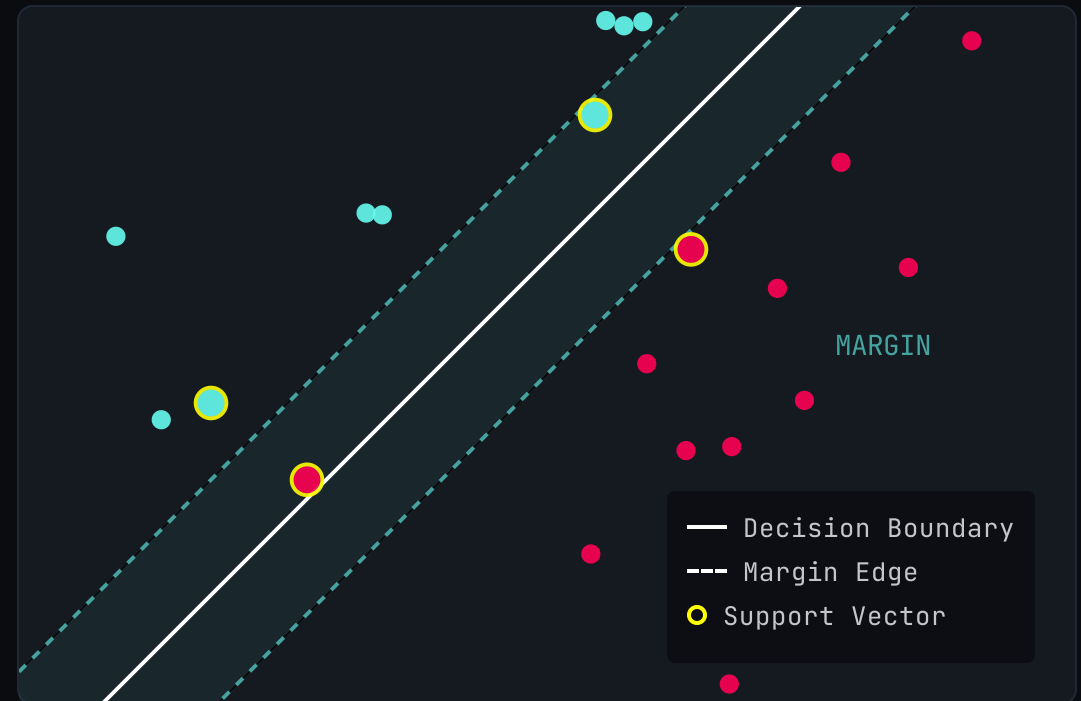
The Margin

The gap or "safety buffer" between the decision boundary and the nearest data points. → Wider margin = More confidence.

COMPONENT 02

Support Vectors

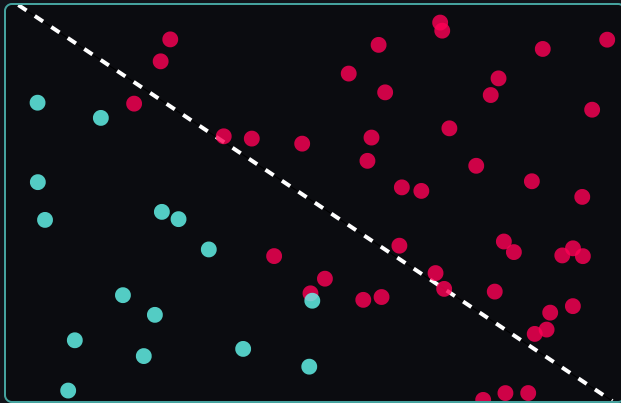
The specific data points that sit on the edge of the margin. They "support" or define the boundary.



SVM Kernels

WHEN A STRAIGHT LINE ISN'T ENOUGH

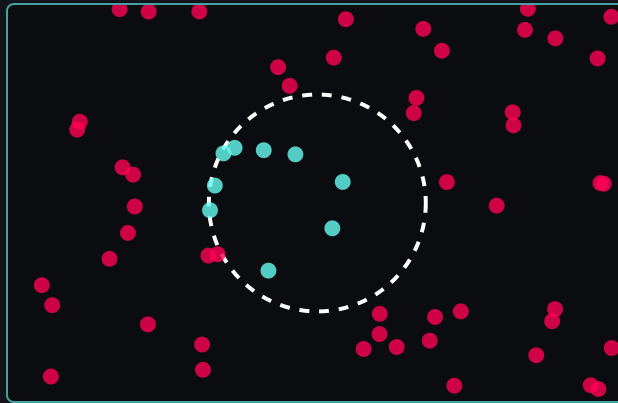
`kernel='linear'`



The Straight Edge

Draws a simple straight line. Best for data that is already clearly separated. Fast and interpretable.

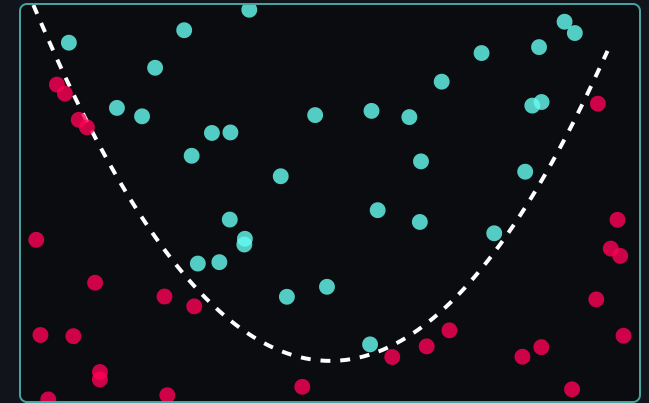
`kernel='rbf'`



Radial Basis Function

Creates curved, flexible boundaries (like islands). The default choice because it handles complex clusters well.

`kernel='poly'`



Polynomial

Draws curved lines based on mathematical degrees. Good for specific geometric patterns but can be slow.

The Classification Report

BEYOND ACCURACY

```
> print(classification_report(y_test, y_pred))
```

```
precision recall f1-score support 0 0.76 0.90 0.83 29 1 0.89 0.75  
0.81 32 accuracy 0.82 61 macro avg 0.82 0.82 0.82 61 weighted avg  
0.83 0.82 0.82 61
```



The Full Story

Accuracy gives you a single grade. The classification report gives you the **report card**.

It breaks down performance by **Class** (0 vs 1) and by **Metric** (Precision vs Recall).

Precision vs. Recall

THE CRITICAL TRADEOFF



Precision

"When the model says YES, how often is it right?"

Focuses on **reliability**. High precision means very few false alarms.

BEST FOR:

Spam Filters (Don't delete important emails!)



Recall

"Of all the actual positives, how many did we find?"

Focuses on **coverage**. High recall means very few missed cases.

BEST FOR:

Cancer Screening (Don't miss a tumor!)



The Tradeoff: Usually, increasing one lowers the other. You must choose what matters more.

The Four-Classifer Showdown

STRENGTHS & WEAKNESSES

ALGORITHM	PROS (STRENGTHS)	CONS (WEAKNESSES)	BEST FOR ...
k-NN	✓ Simple to understand ✓ No training phase	✗ Slow on large data ✗ Sensitive to outliers	<i>Small datasets where similarity matters.</i>
Decision Tree	✓ Highly interpretable ✓ Handles non-linear data	✗ Prone to overfitting ✗ Unstable (high variance)	<i>When you need to explain "Why?"</i>
Logistic Reg.	✓ Fast & efficient ✓ Outputs probabilities	✗ Linear boundaries only ✗ Can't solve XOR problems	<i>Binary classification baselines.</i>

Reading the Results

CONTEXT IS KING



Balanced Data

Metric: Accuracy

When classes are roughly equal (50/50), accuracy is a safe bet. It tells you the overall "grade" of the model.

Example: Predicting Coin Flips



Risk Averse

Metric: Precision

Use this when a **False Alarm** is expensive or annoying. You want to be sure when you say "Yes".

Example: Spam Filters, YouTube Recommendations



Safety Critical

Metric: Recall

Use this when a **Missed Detection** is dangerous. You'd rather have a false alarm than miss a real threat.

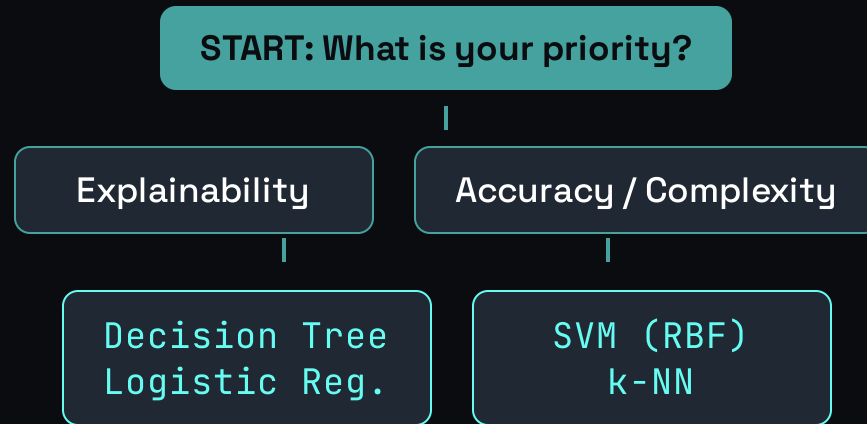
Example: Cancer Diagnosis, Fraud Detection



Pro Tip: Can't decide? Look at the **F1-Score**. It's the harmonic mean (balance) of Precision and Recall.

Choosing the Right Classifier

THE "NO FREE LUNCH" THEOREM



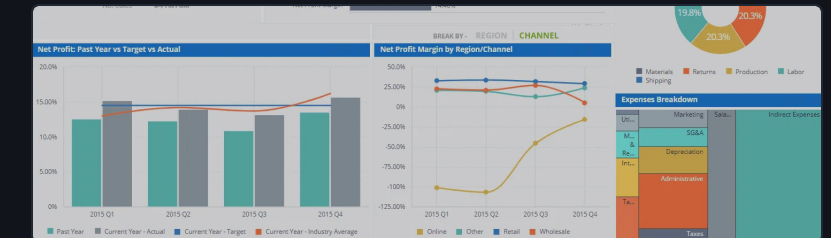
** Also consider: Dataset Size & Linearity*

There is no "Best" Algorithm

The No Free Lunch Theorem states no single model works best for every problem.

You must choose based on:

- Data Size
- Interpretability needs
- Computational cost



Ethics: The COMPAS Case

WHEN "ACCURATE" ISN'T FAIR

Context: COMPAS is an algorithm used in US courts to predict recidivism (likelihood of re-offending) to help judges set bail.

THE METRIC TRAP (ACCURACY)

White Defendants: ~67% **Accurate**

Black Defendants: ~65% **Accurate**

✔ Looks fair, right? The overall accuracy is nearly identical.

THE HIDDEN BIAS (ERRORS)

False Positives (Wrongly Flagged High Risk):

Black vs. White **2x Higher for Black**

False Negatives (Wrongly Flagged Low Risk):

White vs. Black **2x Higher for White**

⚠ The model made different types of mistakes for different groups.

Mission Debrief

CHAPTER 08 COMPLETE

- ✓ **Logistic Regression** isn't regression. It uses the Sigmoid function to output probabilities (0 to 1).
- ✓ **SVMs** maximize the margin (street width) to find the safest boundary. Kernels handle the curves.
- ✓ **Precision vs. Recall** is a tug-of-war. You must choose based on the cost of being wrong.
- ✓ **Ethics Check:** Accuracy can hide bias. Always audit error rates across different groups.

NEXT MISSION

Neural Networks & Deep Learning

```
t.begin_fill()  
  
counter=0  
  
while counter < 4:  
    t.forward(100)  
    t.left(90)  
    counter = counter+1  
  
t.end_fill()  
time.sleep(5)  
|
```